EXP: 07 : 13-08-2026

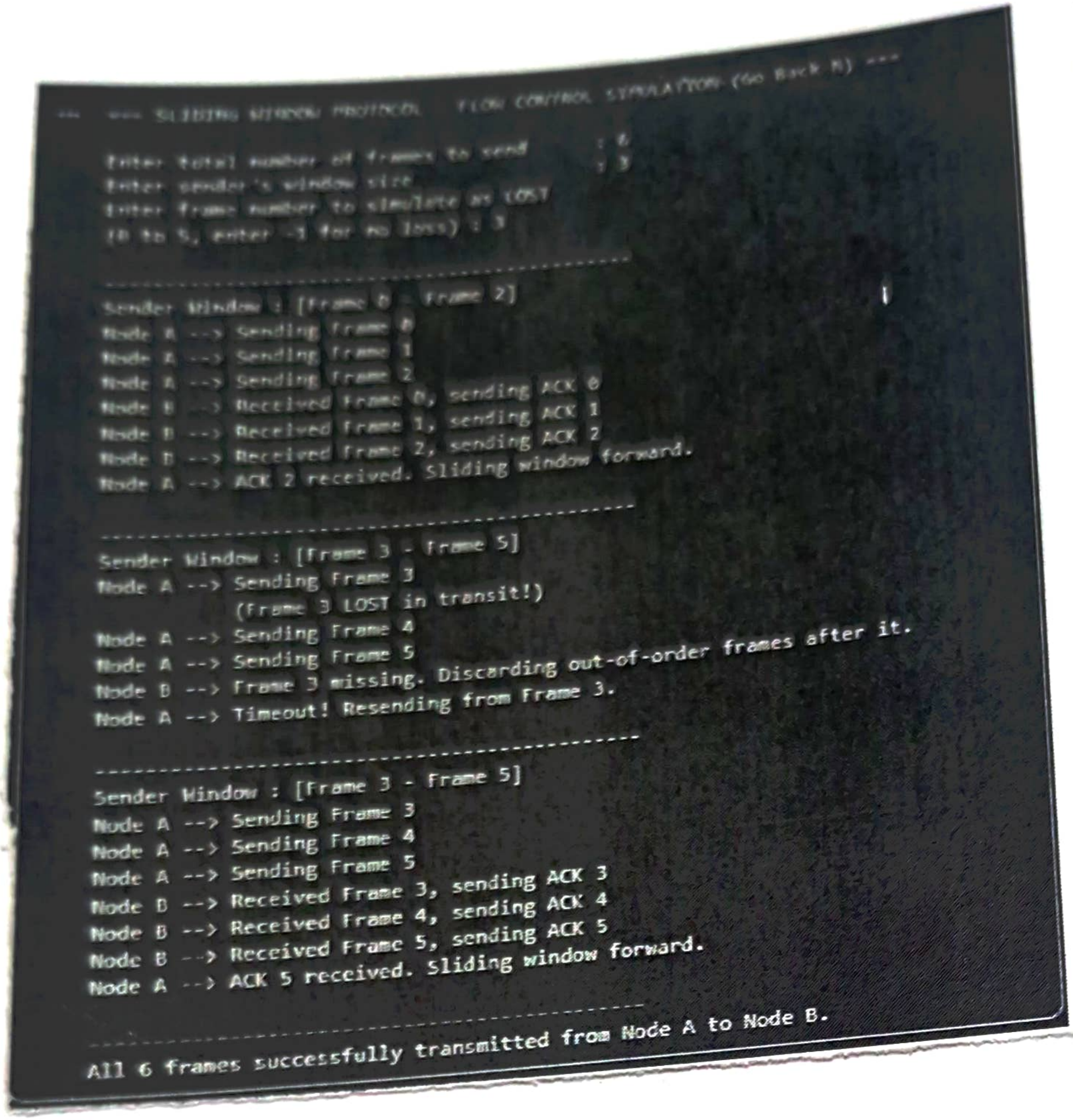
FLOW CONTROL AT CONTROL DATA LINK LAYER

Aim:
Write a Program to implement flow Control at data link layer. Simulat the flow of frames from one node to another

Steps:
1. Input windows size from the user
2. Input a text message from the user
3. Consider 1 character per frame.
4. Creat a frame with following fields [Frae No Data]
5. Send the frames
6. Wait for the acknowledgement form the recevor.
7. Reader a file Called Recivers Buffer.
8. If a Acknowledgement number is as excepted start new set of frames accordingly. If the if NACK is Received, resend the frames accordingly.

Creat a reader file with following feature,

1. Reader a file Called Send Buffer
2. Check the fram no,
3. ~~If~~ the frame with the ACK for the window
4. Show the Program.



```
--  === SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION (Go Back N) ---

  Enter total number of frames to send       : 6
  Enter sender's window size                 : 3
  Enter frame number to simulate as LOST
  (0 to 5, enter -1 for no loss) : 3

  ---------------------------------------------
  Sender Window : [Frame 0 - Frame 2]
  Node A --> Sending Frame 0
  Node A --> Sending Frame 1
  Node A --> Sending Frame 2
  Node B --> Received Frame 0, sending ACK 0
  Node B --> Received Frame 1, sending ACK 1
  Node B --> Received Frame 2, sending ACK 2
  Node A --> ACK 2 received. Sliding window forward.

  ---------------------------------------------
  Sender Window : [Frame 3 - Frame 5]
  Node A --> Sending Frame 3
             (Frame 3 LOST in transit!)
  Node A --> Sending Frame 4
  Node A --> Sending Frame 5
  Node B --> Frame 3 missing. Discarding out-of-order frames after it.
  Node A --> Timeout! Resending from Frame 3.

  ---------------------------------------------
  Sender Window : [Frame 3 - Frame 5]
  Node A --> Sending Frame 3
  Node A --> Sending Frame 4
  Node A --> Sending Frame 5
  Node B --> Received Frame 3, sending ACK 3
  Node B --> Received Frame 4, sending ACK 4
  Node B --> Received Frame 5, sending ACK 5
  Node A --> ACK 5 received. Sliding window forward.

  ---------------------------------------------
  All 6 frames successfully transmitted from Node A to Node B.
```

Program 1

```
def simulate (total_frames, windows_size lost base = 0)
  next seq = 0
 while base < total frames :
  window - end = min (base + windows _size]
  total_frames - 1)
  Print ("/ n- - - - - - - - - -")
  Print (f" Slider window" [frame {base} frame {window_end}]")
   Print (f"Node A sending frame (next next_seq + = 1
    if lost - frame is not None and base
     < = lost _ frame < next _seq )
    Print ("f Node B's Frame {lost frame}
      missing decoding order

for f in range (base , max _ seq ):
  Print (f" f node B-Received frame {f}
  Print (f " Node A-Ack {Next - seq - 1}
    Receiving sliding window forward
   base = next _ seq
 Print ("/n- - - - - - -")
 Print (f" All total_frames} [from frames transmitted from Node
   A to node B")
 def main()
  Print ("== Sliding window Protocol_flow)
  Control (simulating (Go_back _N ) = = \n")
  total_frame : int (input ("Enter total no of frames to send")
  if lost_frame = = -1 ,
   lost _ frame = None.
   simulate ( Total _ frames , windows _ size, lost _ frame
   if _ name _ = = "_main_":
```

Result!

The Program to implement flow control at the data link layer using sliding windows protocol was successfully written in Python. Program executed and verified by simulating the flow of frame between the networks.